

Advanced techniques in applied economics

Lecture 4: Causal discovery and non-Gaussian identification

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Spring 2026

What can we learn from observational data alone?

What changes today?

In Lecture 3 we gave causal meaning to DAGs using factorization, d-separation, and interventions. Today we ask a harder question:

How much of the causal direction can be recovered without intervening?

Constraint-based discovery

Use conditional independence tests to recover the skeleton and the compelled colliders.

Extra identifying structure

Use assumptions such as non-Gaussianity or invariant mechanisms to go beyond Markov equivalence.

Main question: Which arrow directions are visible from conditional independence alone, and which require extra assumptions?

Roadmap for today

1. constraint-based discovery and the PC algorithm
2. why Markov equivalence is the main obstacle
3. why linear Gaussian SEMs do not identify direction
4. LiNGAM and the role of non-Gaussianity
5. more realistic shocks: heteroskedasticity and mean independence
6. a brief link to additive noise models

Conditional independence narrows down the graph; asymmetry assumptions identify direction.

Part 1: Constraint-based discovery

Why conditional independence alone stops at a CPDAG

Constraint-based discovery uses the logic of Lecture 3:

- d-separation implies conditional independence,
- missing edges correspond to some separating set,
- colliders can be oriented from separating sets.

From observational data we may learn

- the **skeleton**,
- the **v-structures**,
- and therefore a **CPDAG**.

What this does *not* buy us

Without extra assumptions, we usually cannot orient every edge.

High-level idea of the PC algorithm

The PC algorithm is the canonical constraint-based discovery method.

Skeleton phase

1. Start from the complete undirected graph.
2. Test marginal and conditional independences.
3. Remove an edge whenever some separating set is found.

Orientation phase

1. Use separating sets to orient v-structures.
2. Apply graphical rules to orient additional compelled edges.

Output

A **CPDAG**, not necessarily a unique DAG.

PC is a discovery procedure based on conditional independence, not on interventions.

The PC algorithm: skeleton phase

Input

Data on $X_1, \dots, X_m$, a conditional independence test, and a significance level α .

Skeleton phase

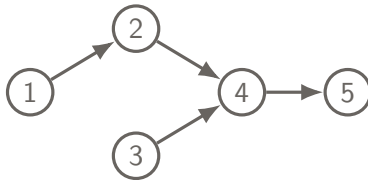
1. Initialize with the complete undirected graph on $\{1, \dots, m\}$.
2. For conditioning sets of size $0, 1, 2, \dots$:
 - ▶ for each adjacent pair (i, j) ,
 - ▶ test $X_i \perp\!\!\!\perp X_j \mid X_S$ for candidate sets S ,
 - ▶ if independence is not rejected, remove edge $i - j$ and store S .

Interpretation

Search for a small set of variables that explains away the apparent association between X_i and X_j .

A toy discovery example

Suppose the true DAG is



Skeleton phase intuition

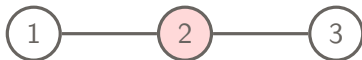
- marginal tests may already remove $1 - 3$ and $2 - 3$,
- then conditioning on 2 may remove $1 - 4$ and $1 - 5$,
- conditioning on 4 may remove $2 - 5$ and $3 - 5$.

Message

The skeleton is found by asking which pairs can be separated by some conditioning set.

Orientation phase: where colliders come from

Suppose the skeleton contains



with no edge between 1 and 3. If the separating set for $(1, 3)$ *does not* contain node 2, then PC orients

$$1 \rightarrow 2 \leftarrow 3.$$

Why?

If 2 were a non-collider, then 2 would have to belong to every minimal separator between 1 and 3.

After orienting all such colliders, PC applies **Meek's rules** to propagate additional arrow directions, while avoiding new v-structures and directed cycles.

A first PC example in R

```
library(pcalg)
library(MASS)
set.seed(1)

Sigma <- matrix(c(1,.7,0,.49,0,
                  .7,1,0,.7,0,
                  0,0,1,.6,0,
                  .49,.7,.6,1,.7,
                  0,0,0,.7,1), 5, 5)

X <- mvrnorm(500, mu = rep(0, 5),
            Sigma = Sigma)
```

```
suffStat <- list(C = cor(X), n = nrow(X))
pc.fit <- pc(suffStat,
            indepTest = gaussCIttest,
            alpha = 0.01,
            labels = paste0("X", 1:5),
            method = "stable")

pc.fit
```

Interpretation

For Gaussian data, the PC algorithm uses partial-correlation tests to recover a CPDAG.

Why economists should be cautious with PC

Constraint-based discovery is elegant, but it rests on demanding assumptions.

Statistical issues

- many CI tests,
- finite-sample error propagation,
- high-dimensional conditioning sets,
- sensitivity to near-unfaithfulness.

Economic issues

- latent confounding is common,
- Gaussianity may be implausible,
- observational data may identify only part of the structure.

PC is best viewed as a disciplined way to learn a CPDAG, not as a button that magically discovers the true causal graph.

Part 2: Why Gaussian linear models are not enough

Why linear Gaussian SEMs do not identify direction

Take two centered Gaussian variables (X, Y) .

Forward representation

$$Y = aX + \varepsilon_Y, \quad \varepsilon_Y \perp\!\!\!\perp X.$$

Backward representation

$$X = bY + \varepsilon_X, \quad \varepsilon_X \perp\!\!\!\perp Y.$$

Key point

The same bivariate Gaussian distribution admits both directions as valid linear SEMs with independent noise.

Gaussianity is convenient for estimation, but it hides causal direction.

What kind of extra assumptions can break the symmetry?

To identify direction from observational data, we need an asymmetry that is invisible in the Gaussian model.

Three common strategies

- **Constraint-based methods:** rely on conditional independence and stop at a CPDAG.
- **Non-Gaussian linear models:** LiNGAM uses non-Gaussianity of the shocks.
- **Nonlinear noise models:** additive noise models use functional asymmetry.

Today's focus

We stay in the linear SEM world and ask what non-Gaussian shocks can reveal.

Part 3: LiNGAM and non-Gaussianity

LiNGAM assumptions

LiNGAM stands for **Linear Non-Gaussian Acyclic Model**.

Model

$$X = BX + \varepsilon,$$

where

- B is compatible with a DAG,
- the errors $\varepsilon_1, \dots, \varepsilon_p$ are mutually independent,
- the errors are non-Gaussian (except possibly one),
- there are no hidden common causes.

Message

LiNGAM keeps linearity and acyclicity, but replaces Gaussian noise by non-Gaussian noise.

Why non-Gaussianity helps

Independence is highly restrictive under linear transformations.

Kac / Darmois–Skitovich intuition

If independent variables remain independent after a nontrivial linear mixing, then the variables must be Gaussian.

Consequence for linear SEMs

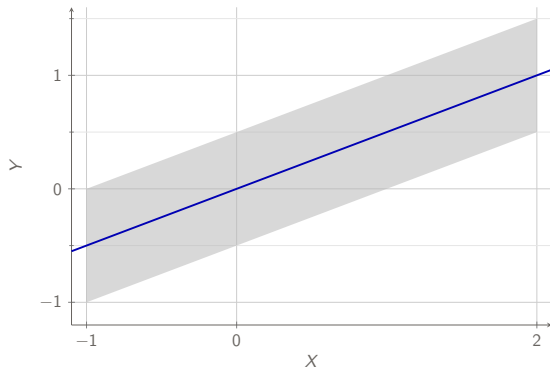
In a non-Gaussian linear SEM, the mixing matrix in

$$X = (I - B)^{-1} \varepsilon$$

is identifiable up to scaling and permutation. With the acyclicity assumption, this is enough to recover the causal ordering and then the edge coefficients.

Non-Gaussianity breaks the symmetry that makes Gaussian linear SEMs directionally ambiguous.

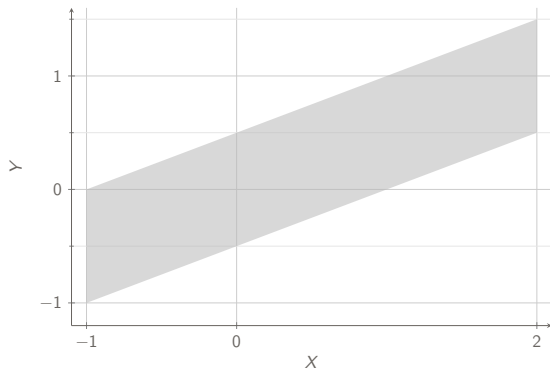
Why non-Gaussianity helps



Consider the forward model: $Y = 0.5X + \epsilon_Y$, $\epsilon_Y \perp\!\!\!\perp X$.

The gray strip is exactly the support implied by $\epsilon_Y \sim U[-\frac{1}{2}, \frac{1}{2}]$ and $X \sim U[-1, 2]$.

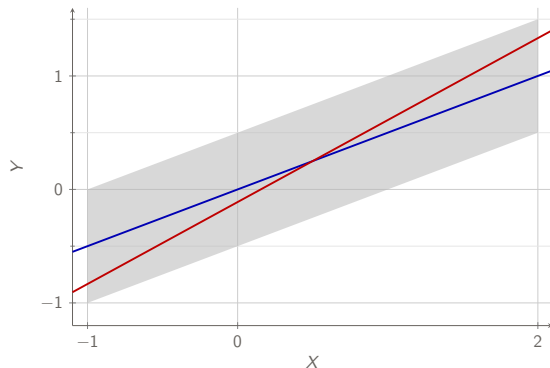
Why non-Gaussianity helps



Question: can the same joint distribution arise from a backward additive-noise model

$$X = bY + c + \varepsilon_X, \quad \varepsilon_X \perp\!\!\!\perp Y ?$$

Why non-Gaussianity helps



Let the red line be the LS fit for predicting X from Y : $(b, c) = \arg \min_{b, c} \mathbb{E}(X - bY - c)^2$.

But then the errors in $X = c + bY + \varepsilon_X$ **could not be independent of Y** .

DirectLiNGAM: the main idea

DirectLiNGAM avoids full ICA and instead looks for an exogenous variable first.

Key insight

If X_j has no parents, then regressing every other variable on X_j leaves residuals that are independent of X_j .

1. Find a variable that is most independent of its regression residuals.
2. Treat it as exogenous and remove its effect from the remaining variables.
3. Repeat on the residual system.

Output

A causal ordering, hence a fully directed DAG in the basic LiNGAM model.

A simple LiNGAM simulation in R

```
set.seed(1)
n <- 2000

x1 <- rt(n, df = 5)
x2 <- 0.8 * x1 + rt(n, df = 5)
x3 <- -0.6 * x1 + 0.7 * x2 +
      rt(n, df = 5)

X <- cbind(x1, x2, x3)
colnames(X) <- c("x1", "x2", "x3")
```

```
# one possible implementation
library(bnlearn)
fit <- direct.lingam(as.data.frame(X))
fit

# older implementations also exist,
# e.g. pcalg::LINGAM(X)
```

Interpretation

Shocks are non-Gaussian, so direction can become identifiable even though the model is linear.

Why this is relevant for economics

Economic shocks are often far from Gaussian.

Examples

- asset-return shocks are heavy-tailed,
- firm-level outcomes are skewed,
- macro shocks often have outliers and asymmetries.

Opportunity

What is a nuisance for Gaussian methods can become useful for identification.

Non-Gaussianity is not only a robustness issue; it can also be an identification resource.

Part 4: Beyond strict independence of shocks

Why independent shocks may be too strong in economics

The basic LiNGAM assumption is that the structural shocks are mutually independent.

Why this may fail

In economic data, shocks often share volatility, exposure, or background uncertainty.

Examples

- common market volatility,
- sectoral exposure to the same aggregate news,
- strategic interaction or equilibrium feedback,
- dependence created by omitted but low-frequency macro conditions.

Consequence

The clean ICA-style separation behind LiNGAM may become fragile or fail outright.

A toy example with common volatility

Consider a linear SEM

$$X_1 = \varepsilon_1, \quad X_2 = aX_1 + \varepsilon_2,$$

but let

$$\varepsilon_1 = \sigma U_1, \quad \varepsilon_2 = \sigma U_2,$$

where U_1, U_2 are independent and non-Gaussian, while the random scale σ is shared.

What happens?

Conditional on σ , the shocks are independent; marginally they are dependent through common volatility.

Economic reading

This is a crude model of common market stress: shocks are not driven by a common mean factor, but their magnitude co-moves across equations.

A mean-independence perspective

A weaker requirement than full independence is often more realistic.

Mean independence

Instead of requiring

$$\varepsilon_i \perp\!\!\!\perp \varepsilon_j \quad (i \neq j),$$

we may only require suitable **mean-independence** conditions. For instance, a noise term may have mean zero given the relevant past or given suitable summary information.

Why this matters

Mean independence allows dependence in higher moments, such as shared volatility, while still preserving enough asymmetry for identification or estimation.

Your extension: a more realistic economic shock model

The motivation is simple:

Classical LiNGAM

Independent non-Gaussian shocks. Elegant, but often too clean for real economic systems.

More realistic alternative

Allow some dependence between shocks, but keep a directional asymmetry through mean independence or related restrictions.

Message

The goal is to keep the identification spirit of LiNGAM while allowing richer and more realistic shock structures.

For economics, the right question is often not are the shocks fully independent? but what weaker asymmetry is still credible?

A toy simulation with common volatility

```
set.seed(1)
n <- 2000

sigma <- abs(rnorm(n)) + 0.3
u1 <- rt(n, df = 5)
u2 <- rt(n, df = 5)

e1 <- sigma * u1
e2 <- sigma * u2
x1 <- e1
x2 <- 0.8 * x1 + e2
```

```
par(mfrow = c(1,2))
plot(x1, x2, pch = 16, cex = .4)
plot(abs(e1), abs(e2), pch = 16,
      cex = .4)
```

Interpretation

The second plot reveals common volatility in the shocks. This violates strict independence, even though the directional SEM remains meaningful.

Part 5: A brief look beyond linearity

Additive noise models

A different route to identifiability is nonlinear asymmetry.

Additive noise model

$$Y = f(X) + \varepsilon, \quad \varepsilon \perp\!\!\!\perp X.$$

Under suitable regularity conditions, the reverse representation

$$X = g(Y) + \tilde{\varepsilon}, \quad \tilde{\varepsilon} \perp\!\!\!\perp Y$$

usually does not exist.

Comparison with LiNGAM

LiNGAM uses linearity plus non-Gaussianity. Additive noise models use nonlinearity plus independent noise.

Where this connects to causal ML

The common theme is the search for an orientation in which the mechanism looks stable and simple.

LiNGAM

Look for a linear model with independent non-Gaussian shocks.

Additive noise / invariance ideas

Look for a direction in which residuals are independent or mechanisms remain stable across environments.

Across many modern methods, causal direction is identified by some form of asymmetry.

Take-away

- Constraint-based discovery uses conditional independence to recover a CPDAG.
- Markov equivalence limits what can be learned from conditional independence alone.
- Linear Gaussian SEMs do not identify direction in general.
- LiNGAM uses non-Gaussianity to restore identifiability.
- For economic data, strict independence of shocks may be unrealistic; weaker asymmetries such as mean independence can be more credible.
- Beyond linearity, additive noise models provide another route to directional identification.

Conditional independence narrows down the graph. Asymmetry assumptions identify direction.

Extension: graphical ideas for time series

From static SEMs to dynamic graphical models

For a multivariate time series, a natural model is

$$X_t = A_1 X_{t-1} + \cdots + A_p X_{t-p} + \varepsilon_t.$$

Directed lagged structure

Zeros in the matrices A_ℓ mean that some lagged variables do not directly predict others. This is the dynamic analogue of missing directed edges.

Contemporaneous structure

If $\varepsilon_t \sim (0, \Sigma_\varepsilon)$, then zeros in

$$K_\varepsilon = \Sigma_\varepsilon^{-1}$$

encode conditional independence among shocks at time t .

Interpretation

A graphical time-series model separates

- **lagged directional effects** from
- **instantaneous conditional dependence of shocks.**

Why economists may care about graphical time series

These ideas are natural in macroeconomics and finance.

Lagged graph

A sparse lagged graph can summarize who predicts whom:

- sectoral spillovers,
- macro transmission channels,
- Granger-causal relations in large VARs.

Shock graph

A sparse graph for ε_t can summarize contemporaneous residual links:

- common financial stress,
- instantaneous market co-movement,
- residual dependence after lagged dynamics are removed.

Dynamic graphical models ask: who predicts whom over time, and which shocks still interact contemporaneously?

A simple workflow in R

A practical strategy is:

1. estimate a sparse VAR for lagged structure,
2. extract residuals,
3. fit a Gaussian graph to the residuals.

```
library(BigVAR)

# X is a T x m matrix of series
mod <- constructModel(X,
                      p = 1,
                      struct = "Basic",
                      gran = c(10, 10),
                      verbose = FALSE)
fit <- cv.BigVAR(mod)
```

```
# residual graph
Ehat <- residuals(fit)
S <- cov(Ehat)

library(glasso)
out <- glasso(S, rho = 0.05)

library(qgraph)
qgraph(-cov2cor(out$wi),
       layout = "spring")
```

Interpretation

The sparse VAR encodes lagged directed effects. The residual precision matrix estimates contemporaneous conditional dependence among shocks.

Looking ahead

Next lecture

We move from observed-variable causal graphs to **latent structures**.

What comes next?

- latent variable models,
- factor models and mixtures,
- latent trees,
- and the bridge from graphical models to neural-network architectures.